

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract

This report evaluates `minimax/minimax-m2.5` under the standard Infron vs OpenRouter Prompt Caching A/B benchmark. OpenRouter leads most cache, cost, throughput, E2E latency, and TTFT outcomes in Throughput First, Price First, and Latency First modes, while Infron leads all five core metrics in TTFT First. The practical interpretation is that OpenRouter was stronger for the general routing modes in this run, while Infron produced the best result when the route objective explicitly prioritized TTFT.

The strict A/B filter retained 767 paired samples and 3068 request-level observations; 66 records were excluded. All core metrics come from response telemetry.

Conclusion Matrix

Routing mode	Cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Throughput First	OpenRouter	OpenRouter	Infron	OpenRouter	OpenRouter
Price First	OpenRouter	OpenRouter	Infron	OpenRouter	OpenRouter
Latency First	Infron	OpenRouter	Infron	OpenRouter	OpenRouter
TTFT First	Infron	Infron	Infron	Infron	Infron

Overall Metrics

Routing mode	Platform	Pairs	Token cache hit rate	Actual cost	Avg throughput	Avg E2E latency	Avg TTFT
Throughput First	Infron	176	95.28%	\$0.24704000	N/A tok/s	4988.85 ms	4123.13 ms
Throughput First	OpenRouter	176	98.51%	\$0.19202371	N/A tok/s	3827.34 ms	3203.80 ms
Price First	Infron	191	96.45%	\$0.27260500	N/A tok/s	8848.37 ms	7428.26 ms
Price First	OpenRouter	191	97.48%	\$0.19811206	N/A tok/s	3091.26 ms	2541.08 ms
Latency First	Infron	200	99.89%	\$0.20062900	N/A tok/s	3399.66 ms	2853.13 ms
Latency First	OpenRouter	200	99.88%	\$0.19451608	N/A tok/s	3224.87 ms	2652.39 ms

Routing mode	Platform	Pairs	Token cache hit rate	Actual cost	Avg throughput	Avg E2E latency	Avg TTFT
TTFT First	Infron	200	99.95%	\$0.19270000	N/A tok/s	3192.90 ms	2714.81 ms
TTFT First	OpenRouter	200	99.53%	\$0.21689979	N/A tok/s	3358.71 ms	2836.95 ms

Method And Reproducibility

The run used the standard 4 groups x 50 rounds design, streaming Chat Completions, platform–default reasoning/thinking behavior, prompt–length tiers `short:1500,medium:8000,long:32000` , and `/v1/chat/completions` only. A/B inclusion allowed up to 50 prompt–token difference within each paired first/second request.

Artifact	Path
Summary JSON	<code>export/minimax_m25_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_summary.json</code>
Paired dataset CSV	<code>export/minimax_m25_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_benchmark_pairs.csv</code>
Request–level JSONL	<code>export/minimax_m25_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_benchmark_requests.jsonl</code>
Filtered records	<code>export/minimax_m25_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_records.json</code>
Excluded–record audit	<code>export/minimax_m25_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_records_excluded.json</code>
Benchmark runner source	<code>scripts/rerun_routing_sort_cache_cost_ab.py</code>
Test source	<code>tests/test_rerun_routing_sort_cache_cost_ab.py</code>
Dataset reference	Built–in <code>business_representative</code> ; request–level export in <code>benchmark_requests.jsonl</code>